

Tahcom AI — Reference Document

Company solution overview (DeskAlerts) & AI engineering concepts. Prepared as a knowledge source for a RAG demo.

1. About Tahcom AI

Tahcom AI (Tahcom) is an artificial intelligence and information technology solutions company. It provides and integrates technology products and AI-enabled solutions for its clients and partners.

2. DeskAlerts — Internal Communication System

DeskAlerts is an internal communication solution offered by Tahcom. It is a notification (alert) system that sends one-way messages to desktop computers and other devices such as tablets and smartphones. The system is secure and easy to use, with multiple functions that suit a range of communication needs, including reporting and customization.

DeskAlerts can be used to:

- Reduce email overload.
- Notify employees instantly in emergencies.
- Send messages that cannot be ignored or missed, and measure communication effectiveness.
- Deliver comprehensive training and educational programs to employees.
- Alert users about outages, information-system maintenance, and system failures.
- Ensure employees read compliance messages.
- Reduce the load on the technical support desk.
- Save time and cost through the ability to respond quickly to problems.

3. AI Engineering Concepts

LLM (Large Language Model): A model trained on large amounts of text that understands and generates language. It only knows general information from its training data and does not know your private documents.

RAG (Retrieval-Augmented Generation): A technique that retrieves relevant information from your own source at question time and lets the model answer from it instead of from general memory. RAG reduces hallucination but does not eliminate it.

Embeddings: Converting text into numeric vectors that represent meaning. Texts with similar meaning get similar vectors, so search works by meaning rather than exact keywords.

Vector Database (e.g., Chroma): Stores embeddings and retrieves the chunk most similar to the question. This is the 'Retrieval' step in RAG.

Chunking: Splitting a document into smaller passages before embedding. Chunk size affects retrieval accuracy.

Prompt Engineering: Carefully crafting the instructions given to the model (role, context, examples, format) to improve its output quality.

Agents: A model that performs steps and calls tools to accomplish a task, rather than only replying. A chatbot replies; an agent gets a task done.

Hallucination & Evaluation: A hallucination is a confident but wrong or fabricated answer. It is reduced with RAG, citation enforcement, and by evaluating answer quality.

AI Engineer vs ML Engineer: An ML engineer builds and trains models and needs strong math and statistics. An AI engineer integrates ready-made models into applications (RAG, LLM APIs) and focuses on shipping products.